CRICKSTATS

Cricket Player Performance Tracking System

A Comprehensive Salesforce Enterprise Solution

Project Documentation & Technical Architecture Report

PROJECT TEAM MEMBERS

CRICKSTATS Development & Implementation Team

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Role: Salesforce Lead Developer & Architect
Specialization in Custom Object Modeling, Salesforce Lightning Components, and Flow Automation.

Pynthamizhan G

Role: Salesforce Admin & Security Specialist
Specialization in Profiles, Permission Sets, Security Configurations, and Roll-up Summaries.

Rakesh R A

Role: Business Analyst & UI/UX Designer
Specialization in Customer Journey Mapping, User Stories, and Lightning App Page Layouts.

Kaviyarasu R

Role: Quality Assurance & Analytics Engineer
Specialization in Functional Testing, Reports, Dashboard Generation, and System Performance Validation.

Project Mentorship & Guidelines: This report outlines the complete lifecycle of the CRICKSTATS Salesforce application developed following Agile methodology and Software Development Life Cycle (SDLC) best practices.

1. Executive Summary & Introduction

1.1 Executive Summary

The **CRICKSTATS** project delivers an enterprise-grade Salesforce application engineered to digitize, centralize, and automate player performance tracking across cricket leagues and tournaments. By eliminating error-prone manual spreadsheets, CRICKSTATS empowers coaches, team managers, and sports analysts with real-time performance analytics, automated statistical roll-ups, and interactive Lightning dashboards.

1.2 Project Overview

CRICKSTATS leverages the robust Salesforce cloud infrastructure to model complex cricket statistics on a match-by-match basis. Built using custom Salesforce objects, Master-Detail relationships, screen flows, and dynamic reporting tools, the platform provides seamless tracking of batting, bowling, and fielding metrics across multiple teams and seasons.

1.3 Purpose & Objectives

The primary objectives of the CRICKSTATS application include:

- **Data Centralization:** Digitizing paper-based and fragmented spreadsheet records into a secure, single source of truth.
- **Automation:** Eliminating manual calculations for metrics such as total runs, wickets, averages, 50s, and 100s using Salesforce Roll-up Summary fields and Formulas.
- **Real-time Analytics:** Delivering instant statistical reports and dynamic Lightning dashboards to evaluate player consistency and team strategy.
- **Process Streamlining:** Enabling quick match performance data entry through intuitive Screen Flows.

1.4 User Story

> *"As a Salesforce Administrator, I want to create a comprehensive player performance tracking application within Salesforce to capture and analyze detailed match-by-match information of cricket players. This application will include custom objects such as Player and Match Performance with carefully defined fields and relationships to ensure data integrity and accessibility. By leveraging master-detail and lookup relationships, the application will enable robust tracking of individual and team performances, facilitate insightful reporting and analytics, and support overall strategic decision-making."*

2. Ideation Phase & Empathy Mapping

2.1 Problem Statements

Traditional cricket management heavily relies on disconnected spreadsheets, paper logbooks, and manual calculations. The following customer problem statements highlight the pain points across key roles:

Persona (Customer)	Goal (I'm trying to...)	Obstacle (But...)	Root Cause (Because...)	Impact (Makes me feel...)
Cricket Team Coach	Track player performance across matches	Cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection
Team Manager	Maintain complete player records	Takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient
Sports Analyst	Analyze player trends and generate reports	Faces difficulty in comparing performances	No centralized analytics system	Confused and dissatisfied

2.2 Empathy Map Canvas Analysis

To ensure the solution effectively addresses user needs, an extensive Empathy Mapping exercise was conducted across target user personas (Coaches, Managers, Analysts, and Salesforce Admins):

- **Think & Feel:** Users feel an urgent need for accurate data and better analytics tools to ensure match selection decisions are driven by hard data rather than guesswork.
- **Hear:** Management and board members constantly request real-time performance reports and data-driven squad justifications.
- **See:** Disconnected spreadsheets, inconsistent performance metrics across different files, and fragmented record-keeping systems.
- **Say & Do:** Spending countless hours manually copying match stats, cross-checking scores, and compiling custom charts.
- **Pain:** Inconsistent stats across reports, high risk of manual entry errors, delayed updates, and lack of historical trend analysis.
- **Gain:** A single centralized analytics tool, instant report generation, automated performance roll-ups, and faster, confident team selection.

2.3 Brainstorming & Concept Selection

During the ideation phase, several architectural approaches were evaluated:

1. *Web-based standalone custom application (PHP/MySQL)* - High maintenance, lacks enterprise security.
2. *Mobile statistics tracking application* - Limited reporting and analytics capabilities.
3. *AI-based standalone performance prediction system* - High development complexity, lack of standard CRM functions.

4. **Salesforce-based Performance Tracking Platform (CRICKSTATS) [SELECTED]:** Provides enterprise security, cloud reliability, drag-and-drop dashboards, low-code automation, and rapid scalability out of the box.

3. Requirement Analysis Phase

3.1 Customer Journey Map

The operational workflow within CRICKSTATS follows a streamlined 6-stage process:

1. **Login:** Secure role-based authentication via Salesforce Lightning UI.
2. **Enter Player Data:** Register player demographic, team, batting, and bowling profiles.
3. **Record Match Performance:** Enter match-by-match metrics (runs, wickets, overs, catches) via guided Screen Flows.
4. **Generate Reports:** System automatically executes background summary roll-ups and triggers reports.
5. **Analyze Results:** View interactive dashboards categorizing run distributions, strike rates, and bowling averages.
6. **Decision Making:** Coaches and managers utilize insights for tactical squad selection.

3.2 Functional Requirements (FRs)

Req ID	Requirement Name	Detailed Description
FR-1	Player Registration	Capture comprehensive player master data including Name, Role, Team, Batting Style, Bowling Style, and Profile Picture.
FR-2	Match Performance Entry	Record individual player performances per match, including Runs, Balls, Fours, Sixes, Overs, Wickets, Maidens, and Catches.
FR-3	Dashboard Generation	Provide dynamic visual representations of player statistics, top run-scorers, and wicket-takers.
FR-4	Report Creation	Allow tabular, summary, and matrix reporting on player trends and team performance metrics.
FR-5	Flow Automation	Automate data filtering and match recording using Salesforce Screen Flows and guided screen components.

3.3 Non-Functional Requirements (NFRs)

- **NFR-1 Usability:** Intuitive interface requiring less than 15 minutes of onboarding. Lightning App provides drag-and-drop navigation.
- **NFR-2 Security:** Role-based access control via Profiles and Permission Sets ensures coaches edit records while analysts generate read-only reports.
- **NFR-3 Reliability:** Zero data loss with cloud redundancy. Transactions are atomic to prevent incomplete match logs.
- **NFR-4 Performance:** Page rendering and report generation for 100+ player records complete in under 2-3 seconds.
- **NFR-5 Availability:** 99.9% uptime powered by Salesforce multi-tenant cloud architecture.

- **NFR-6 Scalability:** System effortlessly scales from single-team management to supporting multi-league tournaments with thousands of players.

4. System Architecture & Solution Design

4.1 Data Flow Diagram (DFD)

The data architecture illustrates the seamless movement of statistics from user interaction to analytical output:

> **[User] → [Salesforce UI / Screen Flow] → [Player Object]** ←(Master-Detail)→ **[Match Performance Object] → [Reports] → [Dashboard]**

4.2 Technology Stack

Architecture Layer	Technology Used	Purpose / Function
Presentation Layer	Salesforce Lightning UI	Delivers modern, responsive interface for desktop and mobile devices.
Application Layer	Salesforce Screen Flows & Business Logic	Executes process automation, input validations, and screen routing.
Data Layer	Custom Objects & Relationships	Stores Player and Match Performance data with strict referential integrity.
Analytics Layer	Salesforce Reports & Dashboards	Generates real-time charts, metrics, and exportable player reports.
Database Layer	Salesforce Multi-Tenant Cloud Database	Provides secure, scalable, and fully backed-up cloud storage.

4.3 Problem-Solution Fit Analysis

Identified Problem	CRICKSTATS Solution	Technical Implementation
Manual, paper-based records	Automated digital entry	Salesforce Lightning custom objects and input forms.
Data loss and corrupted files	Secure Cloud Storage	Salesforce Cloud infrastructure with auto-backups.
Poor analytical capabilities	Interactive Dashboards	Lightning visual components and real-time reports.
Reporting delays	Real-time Roll-ups	Roll-Up Summary fields and dynamic formula fields.

5. Project Planning & Scheduling

5.1 Product Backlog & Sprint Schedule

The development followed Agile Scrum methodology across multiple structured sprints:

Sprint	Epic / FR	Story ID	User Story / Task Description	Points	Priority	Assignee
Sprint-1	Setup	USN-1	Create & configure Salesforce Developer Org environment.	1	High	Kishore Kumar S
Sprint-1	Data Modeling	USN-2	Create custom objects: Player and Match Performance.	3	High	Pynthamizhan G
Sprint-1	Navigation	USN-3	Configure navigation tabs for seamless access.	3	High	Rakesh R A
Sprint-2	UI Design	USN-4	Design dedicated CRICKSTATS Lightning Application.	3	High	Kaviyarasu R
Sprint-2	Data Structure	USN-5	Add custom picklists, formula fields, and text areas.	5	Medium	Kishore Kumar S
Sprint-2	Layouts	USN-6	Customize page layouts and compact layouts.	5	High	Pynthamizhan G
Sprint-3	Automation	USN-7	Implement validation rules and constraints.	3	High	Kaviyarasu R
Sprint-3	Automation	USN-8	Develop Screen Flows for guided player selection & data entry.	5	High	Kishore Kumar S

5.2 Project Velocity & Burndown Tracker

Sprint	Total Points	Duration	Start Date	Planned End Date	Completed Points	Release Date
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	20	05 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	20	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	20	19 Nov 2022

6. Implementation - Data Model & Custom Fields

6.1 Custom Objects Architecture

The system utilizes two primary custom objects linked via a **Master-Detail Relationship**:

1. **Player (__c):** Serves as the Master object containing cumulative career statistics and metadata.
2. **Match Performance (__c):** Serves as the Detail object capturing individual match logs.

6.2 Field Specifications - Player Object

Field Label	API Name	Data Type	Details / Options
Player Name	Player_Name__c	Text (50)	Record Name Field
Player Number	Player_Number__c	Text (3)	Jersey Number
Role	Role__c	Picklist	Batsman, Bowler, Batsman & Wicket Keeper, All Rounder
Batting Style	Batting_Style__c	Picklist	Right-Handed, Left-Handed
Bowling Style	Bowling_Style__c	Picklist	Right-Arm Fast, Left-Arm Fast, Right-Arm Spin, etc.
Team	Team__c	Picklist	CSK, DC, KKR, MI, PBKS, RR, RCB, SRH, GT, LSG
AVG Score	AVG_Score__c	Formula (Number)	Calculated batting average across matches.

6.3 Average Score Formula Logic

The batting average is calculated dynamically using the following formula to prevent division-by-zero errors:

```
IF(
   Matches_Played__c > 0,
   Total_Runs__c / Matches_Played__c,
   0
)
```

6.4 Roll-up Summary Fields on Player Object

Field Label	Roll-up Type	Target Field	Filter Criteria
Matches Played	COUNT	Match Performance	All records included
Number of 50's	COUNT	Match Performance	Runs Scored ≥ 50 AND Runs Scored < 100
Number of 100's	COUNT	Match Performance	Runs Scored ≥ 100 AND Runs Scored < 200
Total Runs	SUM	Runs Scored	All records included
Total 4's / 6's	SUM	Fours / Sixes	All records included
Total Wickets	SUM	Wickets Taken	All records included

7. Process Automation & Screen Flows

7.1 Player Screen Flow Architecture

To enhance user efficiency, a dedicated **Salesforce Screen Flow** was engineered and deployed directly on the application Home Page. This flow allows users to quickly filter players by team and view real-time statistical tables without navigating away.

7.2 Flow Execution Steps

1. **Start Node:** Initiates the Screen Flow upon user access on the Home Page.
2. **Players Screen Component:** Renders a picklist populated from the ***Team__c*** Choice Set.
3. **Get Records Element (Players):** Queries the ***Player__c*** object filtering records where ***Team__c*** = ***dynamic_choice***.
4. **Player Info Screen Component:** Renders a dynamic Data Table featuring player profile attributes:
   - Profile Picture & Player Name
   - Matches Played & Total Runs
   - AVG Score (Formula)
   - Batting & Bowling Styles
   - Number of 50s & 100s
5. **End Node:** Finalizes transaction and refreshes view.

7.3 Home Page Integration

The flow was added as a primary Lightning Component on the CRICKSTATS custom Home Page layout. App activation was set as the default for the CRICKSTATS application for System Administrators, Coaches, and Managers.

> **Automation Key Advantage:** Eliminates manual report filtering by enabling instant single-click team performance views directly on the landing dashboard.

8. Testing, Verification & Analysis

8.1 Functional & Performance Testing

Comprehensive testing was conducted within the Salesforce Developer sandbox environment to validate business logic and system response times:

- **Data Integrity Testing:** Verified that deleting a Match Performance record automatically updates the parent Player's Roll-up Summary fields (Total Runs, Matches Played, Wickets).
- **Validation Rules:** Tested constraints ensuring negative runs or overs cannot be submitted.
- **Formula Precision:** Verified that division by zero during initial player creation (0 matches) gracefully returns 0.00 average score.
- **System Performance:** Monitored query times during multi-record roll-up updates; dashboard rendering completed within 1.8 seconds.

8.2 Advantages & Disadvantages

Key Advantages	System Limitations / Constraints
• Centralized cloud repository for all team data. • Real-time visual dashboards and analytics. • Automated roll-ups reduce manual updates by 90%. • Role-based enterprise security and field-level controls. • Highly scalable for future league expansions.	• Requires active Salesforce user licenses. • Dependent on active internet connectivity. • Limited offline editing native capabilities.

9. Conclusion & Future Roadmap

9.1 Conclusion

The **CRICKSTATS** application successfully digitizes and modernizes cricket performance management through the power of Salesforce CRM. By replacing manual, error-prone tracking systems with structured custom objects, automated roll-up summaries, and interactive Screen Flows, CRICKSTATS empowers sports organizations to make fast, accurate, data-driven decisions. The project demonstrates a rigorous implementation of SDLC principles, Agile execution, and cloud architecture standards.

9.2 Future Scope & Enhancements

To further extend the capabilities of CRICKSTATS, the following future roadmap is planned:

- **Salesforce Mobile Integration:** Deploying custom Lightning Web Components (LWC) optimized for touch interfaces during live matches on sidelines.
- **Einstein AI Performance Prediction:** Leveraging Salesforce Einstein AI to analyze historical pitch conditions, bowler matchups, and predict player success rates.
- **Video Analytics Integration:** Linking match timestamps with video clips to allow biomechanical stroke analysis directly from player records.
- **Global Ranking Engine:** Automated ranking system calculating player ratings based on ICC-style weightage algorithms.
- **Public Community Portal:** Utilizing Salesforce Experience Cloud to publish fan-facing statistics portals and tournament leaderboards.

9.3 Appendix & Project References

- Salesforce Developer Documentation & Lightning App Builder Guide.
- Salesforce Flow Builder & Roll-up Summary Best Practices.
- Agile Project Management Backlog & Velocity Burndown Models.

***** THANK YOU *****